SIMATS ENGINEERING

ASSESSMENT TOOL 2

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Course Name: Reinforcement Learning for Environment and Agent
Course Code: MLA03

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Code of Conduct:

I M.S.Dinesh 192424016 certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.S.Dinesh

1. RL-Based Warehouse Robot

State: Robot location, obstacle positions, battery level, item location, destination, and warehouse map.

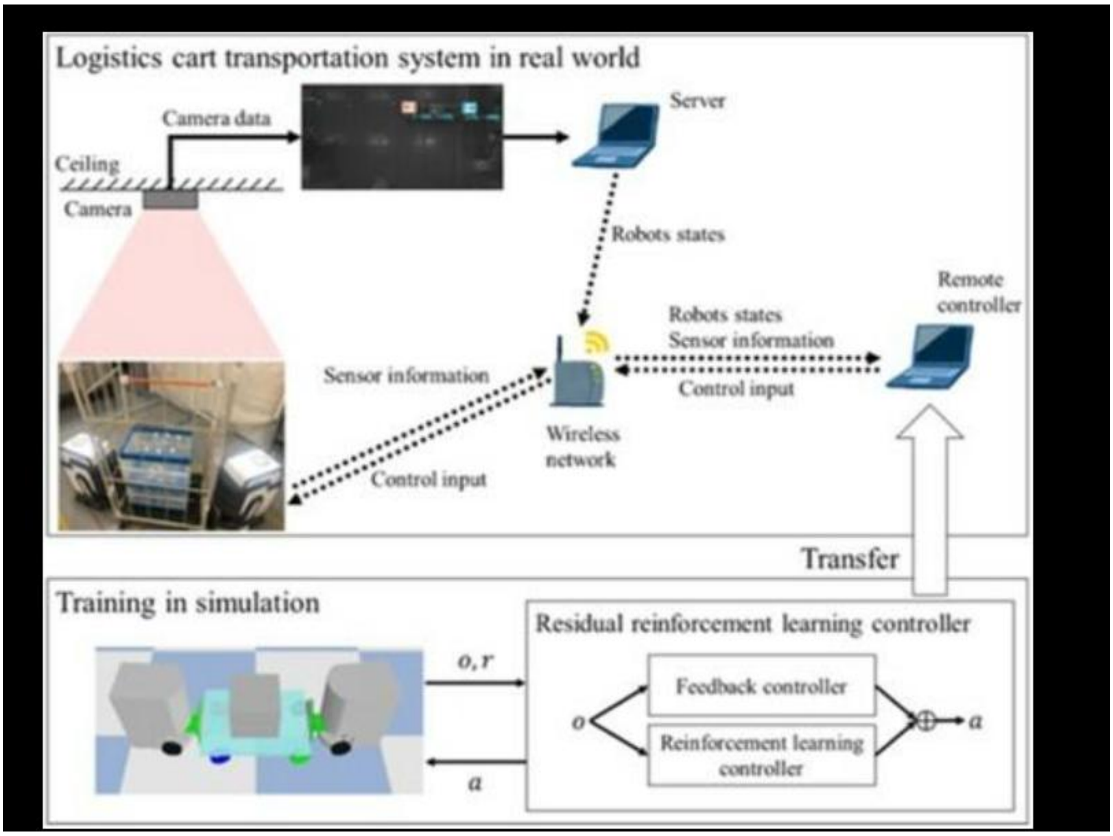
Action: Move forward, backward, left, right, pick item, place item, recharge battery, and avoid obstacles.

Reward Function: +20 for successful delivery, +10 for correct item pickup, -10 for collision, -5 for delay, -15 for wrong delivery, and +5 for energy-efficient movement.

Policy: The robot selects actions that maximize long-term cumulative rewards while completing warehouse tasks efficiently.

Environment: Warehouse containing shelves, robots, workers, obstacles, and delivery locations.

Analysis: The reward function encourages faster deliveries, accurate item handling, and safe navigation. High rewards for successful deliveries improve efficiency, while penalties reduce unsafe behavior and unnecessary delays. A balanced reward function enables the robot to learn an optimal policy that maximizes productivity and minimizes accidents.



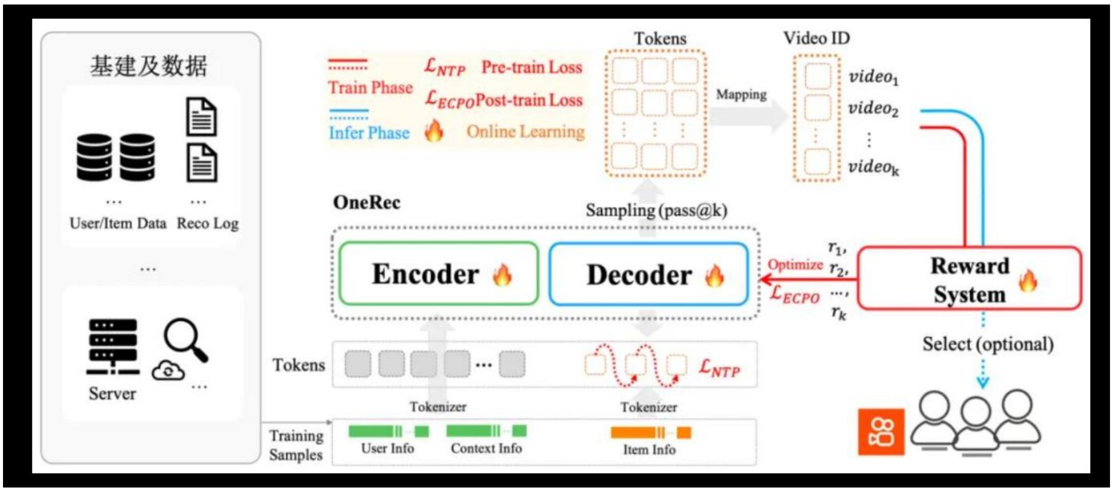
2. Movie Recommendation System

State: User profile, watch history, ratings, preferred genres, viewing time, and recently watched movies.

Action: Recommend a movie, TV series, or documentary.

Reward Function: +10 for watching the recommended content, +5 for completing the movie, +3 for giving a positive rating, and -5 for skipping the recommendation.

Analysis: Positive rewards improve recommendation quality and increase user satisfaction. Exploration introduces new content, while exploitation recommends previously successful genres. A balance between both strategies prevents repetitive recommendations and improves long-term user engagement.



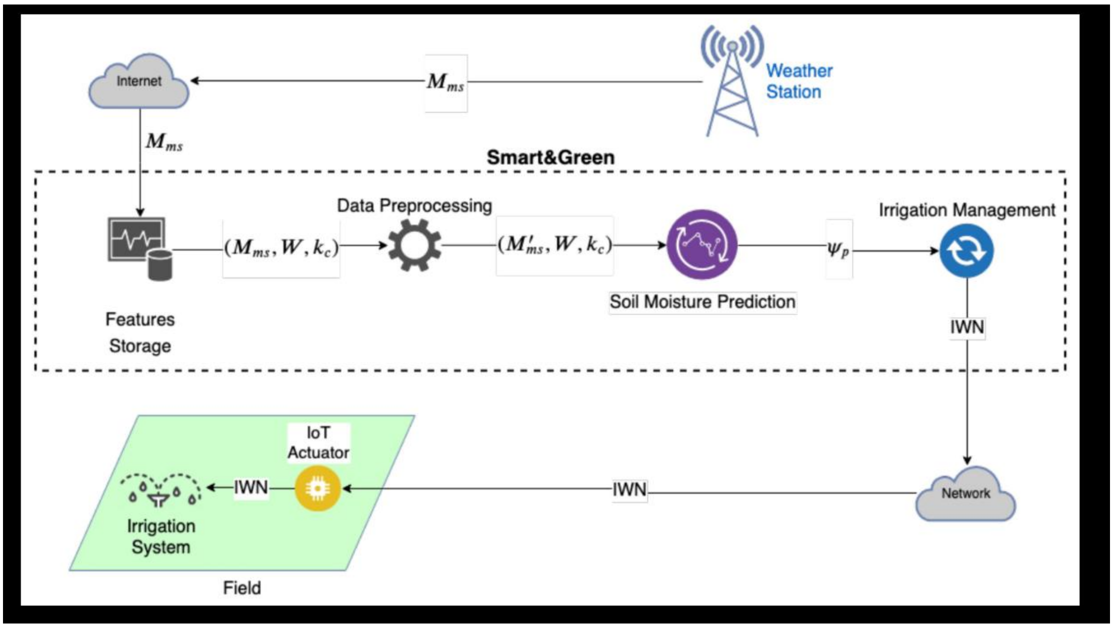
3. Smart Agriculture System

State: Soil moisture, temperature, humidity, rainfall prediction, fertilizer level, and crop growth stage.

Action: Increase irrigation, reduce irrigation, apply fertilizer, delay watering, or maintain the current schedule.

Reward Function: +20 for healthy crop growth, +15 for water conservation, +10 for optimal fertilizer use, and -10 for water wastage or crop damage.

Analysis: Crop growth depends on long-term decisions, making rewards delayed. Reinforcement Learning algorithms such as Q-Learning are suitable because they learn optimal policies by maximizing future cumulative rewards and improving agricultural productivity.



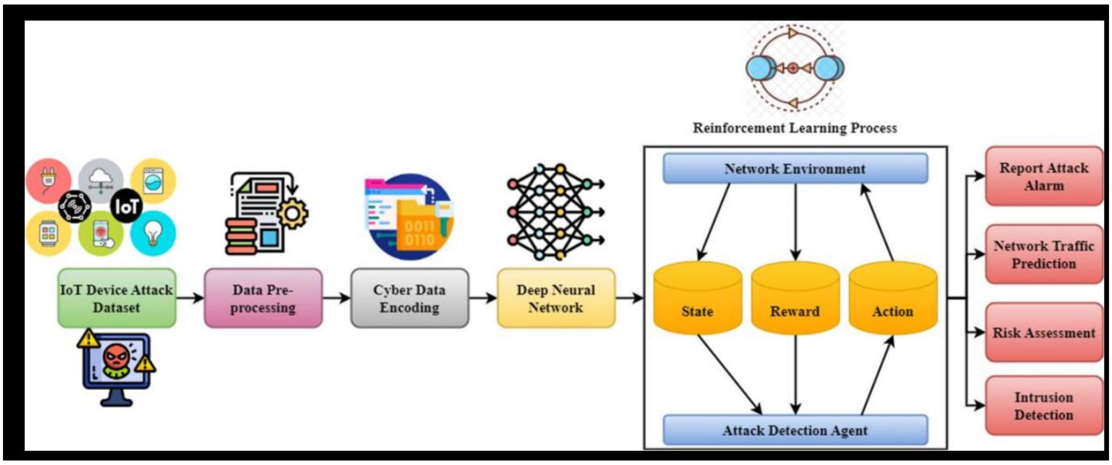
4. Cybersecurity Intrusion Detection

State: Network traffic, login attempts, packet size, protocol type, source IP, and detected threats.

Action: Allow traffic, block connection, isolate device, generate alert, or request authentication.

Reward Function: +20 for detecting attacks, +10 for blocking malicious traffic, -15 for false alarms, and -20 for allowing attacks.

Analysis: Sparse rewards occur because attacks are infrequent. Real-time decision-making requires fast and accurate responses. Proper reward design improves detection accuracy while reducing false positives and maintaining network security.



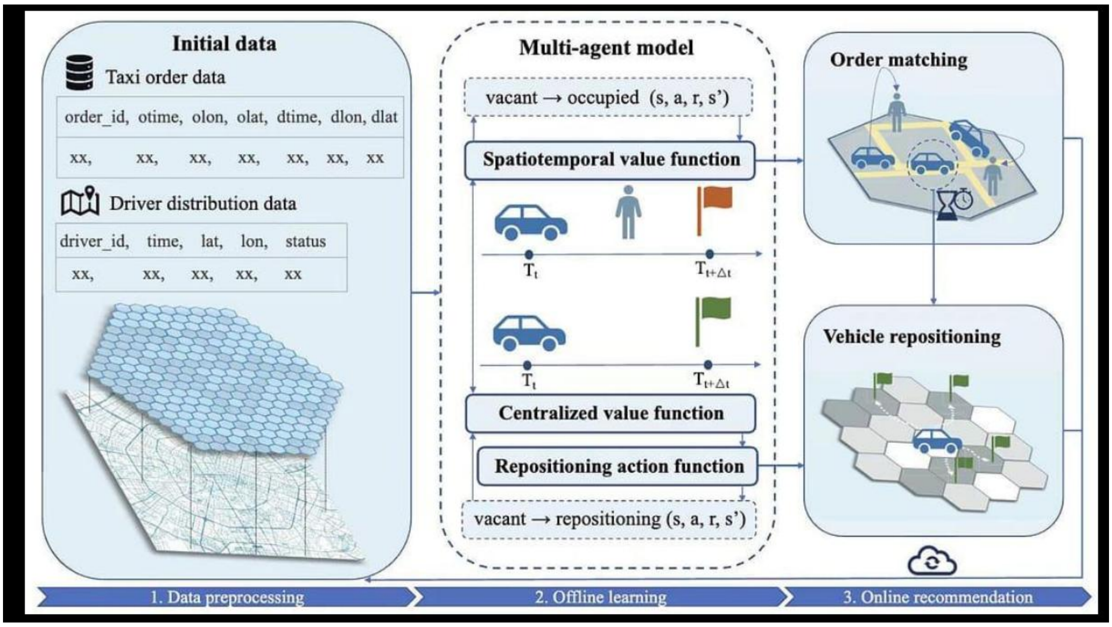
5. Ride-Sharing Optimization

State: Driver locations, passenger requests, traffic conditions, demand level, weather, and current pricing.

Action: Assign drivers, increase or decrease fare, reposition drivers, or reject requests.

Reward Function: +20 for successful ride completion, +10 for reduced waiting time, +5 for balanced driver utilization, and -10 for long delays or ride cancellations.

Analysis: Traffic and demand create environmental uncertainty. The RL agent continuously adapts to changing conditions. Ethical concerns include unfair surge pricing, customer dissatisfaction, and equal service availability. Fair reward design helps balance profit with customer satisfaction.



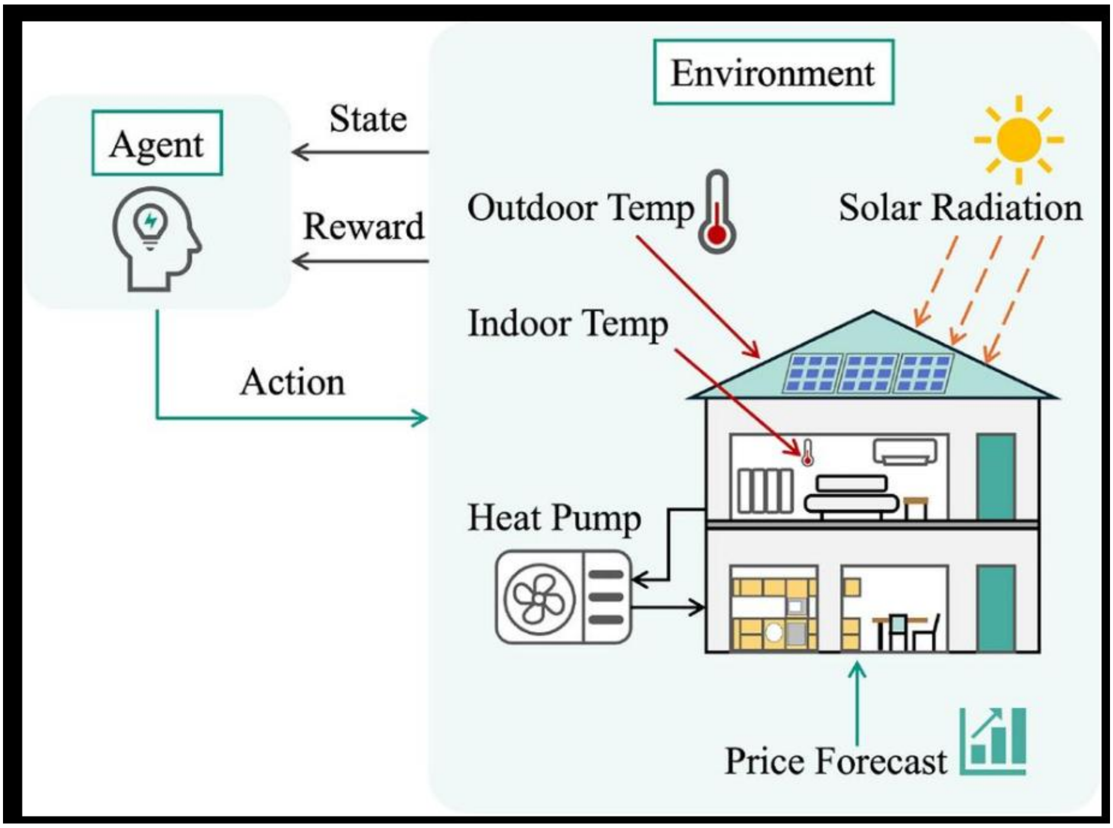
6. Smart Home Automation

State: Room temperature, occupancy, weather conditions, lighting level, energy usage, and time of day.

Action: Turn lights on or off, adjust heating, adjust cooling, and change appliance settings.

Reward Function: +15 for user comfort, +10 for energy savings, -10 for unnecessary energy consumption, and -5 for discomfort.

Analysis: Comfort and energy efficiency are conflicting objectives. Reward shaping assigns appropriate weights to both goals, allowing the system to provide comfort while minimizing electricity consumption.



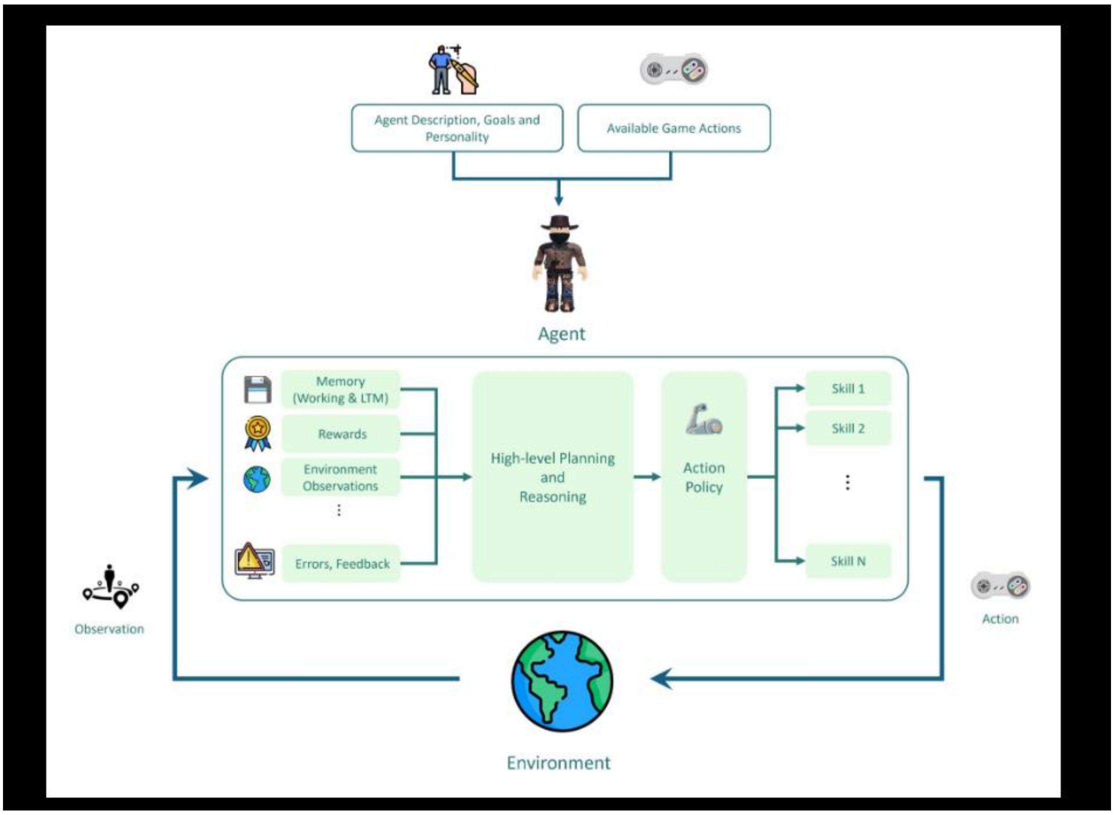
7. RL-Based Strategy Game Agent

State: Current game board, available resources, player position, opponent status, and remaining turns.

Action: Attack, defend, collect resources, build structures, or move units.

Reward Function: +50 for winning the game, +20 for completing objectives, +10 for resource collection, and -20 for losing important units.

Analysis: The strategy game is modeled as a Markov Decision Process where each state depends only on the current state and action. Long-term rewards encourage strategic planning. Exploration discovers new winning strategies, while exploitation applies previously successful actions for consistent performance.



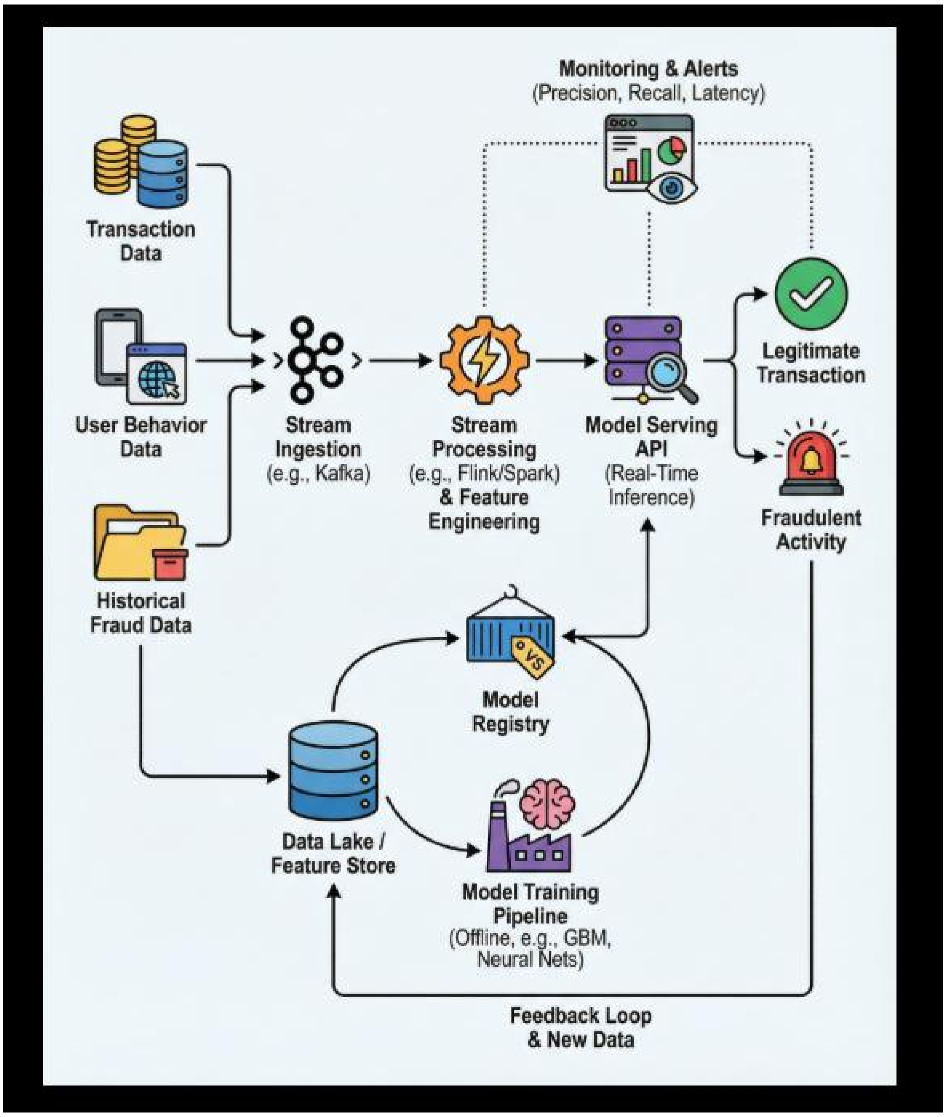
8. Fraud Detection System

State: Transaction amount, account history, transaction location, device information, and transaction frequency.

Action: Approve transaction, reject transaction, or request additional verification.

Reward Function: +20 for correctly detecting fraud, +10 for approving genuine transactions, -20 for missed fraud, and -10 for false rejection.

Analysis: Fraud cases are highly imbalanced and feedback is often delayed. Incorrect decisions can result in financial loss or customer inconvenience. Reinforcement Learning continuously improves fraud detection by learning from previous outcomes.



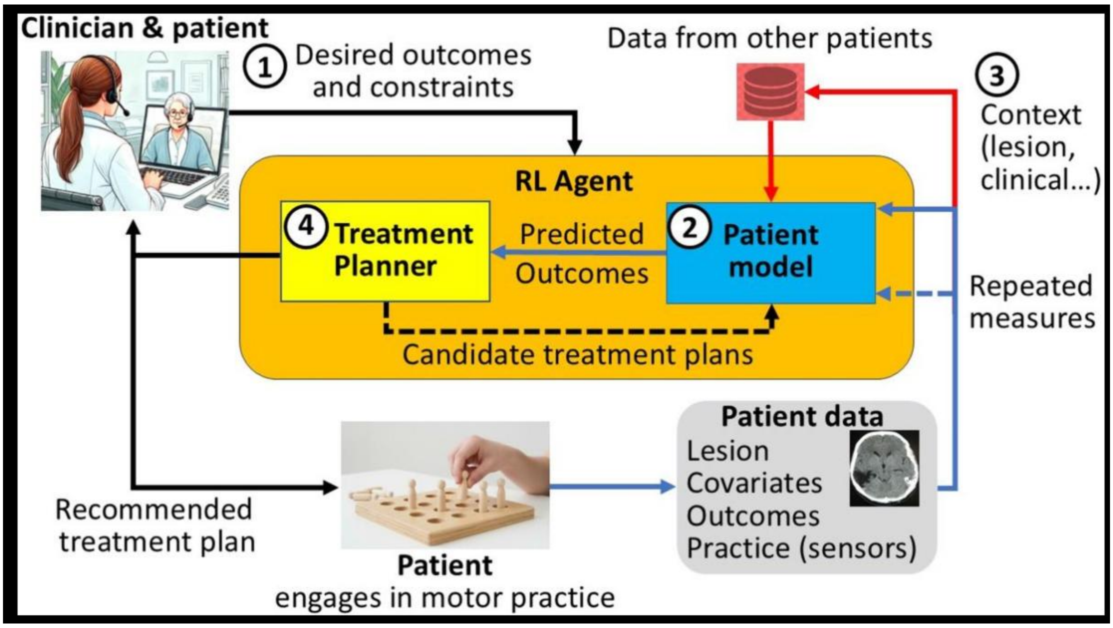
9. Personalized Rehabilitation System

State: Patient progress, exercise history, pain level, mobility score, heart rate, and recovery stage.

Action: Recommend easy, moderate, or advanced rehabilitation exercises.

Reward Function: +20 for improved recovery, +15 for exercise completion, +10 for patient satisfaction, and -15 for injury or excessive pain.

Analysis: Recovery improvements occur gradually, leading to delayed rewards. Patient conditions vary significantly, requiring personalized policies. Ethical considerations include patient safety, fairness, privacy protection, and medical supervision throughout the rehabilitation process.



10. Smart Grid Electricity Management

State: Electricity demand, power generation, battery storage level, renewable energy availability, and grid load.

Action: Distribute electricity, store energy, release stored energy, adjust power generation, and reduce peak demand.

Reward Function: +20 for stable electricity supply, +15 for reducing peak demand, +10 for efficient renewable energy utilization, and -20 for power failures or overload.

Analysis: The Smart Grid is modeled as a Markov Decision Process consisting of states, actions, rewards, policy, and environment. The reward function balances efficiency and reliability. Challenges include large-scale systems, real-time decision-making, and handling continuously changing electricity demand. Reinforcement Learning enables efficient energy distribution while maintaining grid stability.